

Elijah Pivo

Curriculum Vitae

MIT Sloan School of Management
Operations Research and Statistics
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EDUCATION

- 2026 PhD, Social and Engineering Systems
Massachusetts Institute of Technology
Institute for Data, Systems, and Society
Dissertation: *A Holistic Approach for the Simulation, Evaluation, and Design of Organ Transplantation Policies in the U.S.*
Advisor: Dimitris Bertsimas
- 2018 BS, Electrical Engineering and Computer Engineering
Johns Hopkins University

PROFESSIONAL APPOINTMENTS

- 2026 – Present Postdoctoral Associate
MIT Sloan School of Management
- 2018 – 2019 Robotics Engineer
Honeybee Robotics

PUBLICATIONS

Refereed Journal Articles

- 2025 **Elijah Pivo**, James Alcorn, Dimitris Bertsimas, Sarah E Booker, Keighly Bradbrook, Thomas G Dolan, Lindsay V Larin, Kayla R Temple, and Nikolaos Trichakis. “Modernizing the Design Process for US Organ Allocation Policy: Toward a Continuous Distribution Policy for Kidneys.” *Transplantation Direct*, 11(9), e1780.
- 2022 Richard Pates, Andres Ferragut, **Elijah Pivo**, Pencheng You, Fernando Paganini, and Enrique Mallada. “Respect the Unstable: Delays and Saturation in Contact Tracing for Disease Control.” *SIAM Journal on Control and Optimization*, 60(2), S196–S220.

Manuscripts in Preparation

- 2026 **Elijah Pivo**, Thomas Athey, Nikolaos Trichakis, and Dimitris Bertsimas. "Addressing Geographic Access Disparities Under Time-Sensitive Logistics in Kidney Allocation Policy." In preparation for *Management Science*.
- 2026 Thomas Athey, **Elijah Pivo**, Matthew Hartwig, Marie Budev, Dennis Lyu, Kaitlin Swanner, Keighly Bradbrook, Nikolaos Trichakis, and Dimitris Bertsimas. "Optimization-Based Design of Organ Allocation Policy: Mitigating Biologic Access Disparities in Lung Transplantation." In preparation for *American Journal of Transplantation*.

Grants in Preparation

- 2026 "NSF TTP-P: A National Platform for Simulation-Driven Design of U.S. Organ Allocation Policy." Proposal in preparation for submission to the NSF Translation to Practice program. In partnership with UNOS; collaborators include Dimitris Bertsimas and Kaitlin Swanner.

Published Conference Abstracts and Conference Proceedings

- 2026 **Elijah Pivo**, Thomas Athey, Maura Hegarty, Nikolaos Trichakis, and Dimitris Bertsimas. "A Unified, Fast, and Transparent System for Designing Organ Allocation Policies." *American Journal of Transplantation* 26(7), S257. Abstract Only.
- 2026 Thomas Athey, **Elijah Pivo**, and Dimitris Bertsimas. "Lung Allocation Policy Optimization to Equalize Transplant Access Across Blood Types." *American Journal of Transplantation*, 26(7), S1028. Abstract Only.
- 2023 **Elijah Pivo**, Nikolaos Trichakis, Dimitris Bertsimas, Sarah Booker, Keighly Bradbrook, Joanne White, Lindsay Larkin, Thomas Dolan, and James Alcorn. "Powerful New Methods and Tools for Designing Allocation Policy for Deceased-Donor Organs." *American Journal of Transplantation*, 23(6), S474-S474. Abstract Only.
- 2016 Chris Assad, Javier Rodriguez, **Elijah Pivo**, Calvin Huang, Michael Pomerantz, and Adrian Stoica. "Live Demonstration: BioSleeve, a Wearable Hands-Free Gesture Control Interface." *IEEE Sensors*, Orlando, FL, USA, 30 Oct–3 Nov.

AWARDS AND HONORS

- 2018 William H. Huggins Award, Electrical and Computer Engineering, Johns Hopkins University
- 2017 Tau Beta Pi Engineering Honor Society, Elected to Membership as a Junior

GRANTS AND FELLOWSHIPS

- 2025 MIT HEALS Graduate Fellowship
- 2020 U.S. National Science Foundation Graduate Research Fellowship Program
- 2020 MIT IDSS Michael Hammer Fellowship

INVITED TALKS

- 2024 “AI and Policy Development,” *The Organ Donation and Transplantation Alliance National Critical Issues Forum*, Orlando, FL, 6 Sep.

CONFERENCE PARTICIPATION

Sessions Organized

- 2026 “Optimization for Societal Systems,” *INFORMS Optimization Society Conference*, Atlanta, GA, 20–22 Mar.

Presentations

- 2026 “Advancing the Simulation, Evaluation, and Design of Organ Transplantation Policies in the U.S.,” *INFORMS Annual Meeting*, San Francisco, CA, 1 Nov.
- 2026 “Modernizing the Development of Organ Allocation Policy,” *INFORMS Healthcare Conference*, Raleigh, NC, 28 Jul.
- 2026 “A Unified, Fast, and Transparent System for Designing Organ Allocation Policies,” *American Transplant Congress*, Boston, MA, 22 Jun.
- 2026 “Advancing the Design of U.S. Organ Transplant Policy with Optimization,” *INFORMS Optimization Society Conference*, Atlanta, GA, 21 Mar.
- 2025 “Solving for Welfare, Fairness, and Feasibility: Modern Computational Methods for the U.S. Organ Transplant System,” *INFORMS Annual Meeting*, Atlanta, GA, 26 Oct.
- 2024 “Thousands Unused: The Problem of Lossy Allocation in the National Kidney Transplant System and Opportunities for Operations Research,” *INFORMS Annual Meeting*, Seattle, WA, 23 Oct.
- 2023 “Powerful New Methods and Tools for Designing Allocation Policy for Deceased-Donor Organs,” *American Transplant Congress*, San Diego, CA, 5 Jun.
- 2023 “Bringing Modern Analytics to Kidney Transplantation Policy Design,” *INFORMS Annual Meeting*, Phoenix, AZ, 15 Oct.
- 2022 “An Inverse Optimization Approach to Causal Inference,” *INFORMS Annual Meeting*, Indianapolis, IN, Analytics and AI, 16 Oct.

CAMPUS AND DEPARTMENTAL TALKS

- 2026 Panelist, "Data-Driven Decision Making," *SES10: 10th Anniversary Symposium of the SES Doctoral Program*, Cambridge, MA, 9 Oct.
- 2026 "Operations Research for the U.S. Organ Transplant System," *MIT HEALS Annual Symposium*, Cambridge, MA, 16 Mar.
- 2026 "Operations Research for the U.S. Organ Transplant System: Kidney Transportation Logistics," *Holistic AI for Medicine*, Cambridge, MA, 25 Feb.

TEACHING EXPERIENCE

Massachusetts Institute of Technology

Optimization Methods (Fall 2024)

Analytics for a Better World (Spring 2022)

Johns Hopkins University

Control Systems (Spring 2020)

EXTRA TRAINING

- 2024 Massachusetts Institute of Technology Kaufman Teaching Certificate

COMMUNITY INVOLVEMENT AND OUTREACH

- 2020 STEM Achievement in Baltimore Elementary Schools Instructor

RELATED PROFESSIONAL SKILLS

Programming & Computing: Julia, Python, C/C++, Java, MATLAB, Git, Linux, Slurm

Optimization: JuMP, Gurobi, MOSEK, Knitro

Web & Databases: Vue.js, AWS, SQL, SQLite

Embedded Systems: PCB Design & Fabrication (KiCad), FPGA Programming (VHDL), Microcontrollers

Engineering & Robotics: SolidWorks, SysML, ROS

PROFESSIONAL MEMBERSHIPS

Institute for Operations Research and the Management Sciences (INFORMS),
Member

REFERENCES

Dimitris Bertsimas, PhD

Boeing Leaders for Global Operations Professor of Management
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MIT Sloan School of Management
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